Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

The FEDVR basis - 1 - basic properties

- From Ohmi *et al*, Phys, Rev. A 92 (2015), we get

$$\pi_i^s(r) = \frac{1}{\sqrt{\omega_i}} \prod_{\substack{k=1\\k\neq i}}^{N_p} \frac{r - r_k^s}{r_i^s - r_k^s}, \qquad \pi_i^s(r_k^s) = \delta_{ik}/\sqrt{\omega_i^s}, \tag{1}$$

$$\left.\frac{d\pi_i^s(r)}{dr}\right|_{r=r_j^s} = \frac{1}{\sqrt{\omega_i}} \begin{cases} \frac{1}{r_i^s - r_j^s} \prod_{\substack{k=1\\k\neq i,j}}^{N_p} \frac{r_j^s - r_k^s}{r_i^s - r_k^s}, & i \neq j, \\ \frac{1}{2}\left(\left[\pi_i^s(r_{N_p}^s)\right]^2 - \left[\pi_i^s(r_1^s)\right]^2\right), & i = j, \end{cases} \tag{2}$$

- N_s sectors $r_0 \leqslant \ldots r_s \leqslant r_{N_s}$ and N_p nodes a sector, which makes the number of point N so

$$N = N_s(N_p - 1) + 1 \tag{3}$$

$$n = (s-1)(N_p - 1) + i \tag{4}$$

with

$$i = 1, \ldots, N_p, \qquad s = 1, \ldots, N_s, \qquad n = 1, \ldots, N$$

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The FEDVR basis - 2 - The global basis

- The global FEDVR basis is defined, for $n = (s-1)(N_p-1)+N_p$, with $s = 1, ..., N_s - 1$

$$\Pi_n(r) = \frac{1}{\sqrt{\Omega_n}} \begin{cases} \sqrt{\omega^s_{N_p}}\pi^s_{N_p}(r), & r \in [r_{s-1}, r_s], \\ \sqrt{\omega^{s+1}_1}\pi^{s+1}_1(r), & r \in [r_s, r_{s+1}], \end{cases} \qquad \Omega_n = \omega^s_{N_p} + \omega^{s+1}_1, \tag{5}$$

  This function bridges the sectors s and $s+1$ and is continuous through the boundary between them (but its derivative is not).

- For the other values of n corresponding to the quadrature points inside sectors, i.e.

$$n = (s-1)(N_p-1)+i \tag{6}$$

  and at the outer boundary $r_N = r_m$ of the box, we have

$$\Pi_n(r) = \pi^s_i(r), \qquad r \in [r_s, r_{s+1}], \qquad \Omega_n = \omega^s_i, \tag{7}$$

$$\Pi_n(r_{n'}) = \delta_{nn'}/\sqrt{\Omega_n}, \tag{8}$$

$$\int_0^{r_m} \Pi_n(r) f(r) \Pi_{n'}(r) dr \sim f(r_n)\delta_{nn'}. \tag{9}$$

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The FEDVR basis - 3 - the derivative

- We can extend Eq. (2) to any r of the domain and write

$$\frac{d\pi_i^s(r)}{dr} = \frac{1}{\sqrt{\omega_i^s}} \sum_{\substack{j=1 \\ j\neq i}}^{N_p} \frac{1}{r_i^s - r_j^s} \prod_{\substack{k=1 \\ k\neq j,i}}^{N_p} \frac{r - r_k^s}{r_i^s - r_k^s}, \tag{10}$$

- In terms of Π_n, the derivative is expressed as, for $n = (s-1)(N_p - 1) + N_p$, with $\Omega_n = \omega_{N_p}^s + \omega_1^{s+1}$

$$\frac{d\Pi_n(r)}{dr} = \frac{1}{\sqrt{\Omega_n}} \begin{cases} \sqrt{\omega_{N_p}^s}\, d\pi_{N_p}^s(r)/dr, & r \leqslant r_s, \\ \sqrt{\omega_1^{s+1}}\, d\pi_1^{s+1}(r)/dr, & r \geqslant r_s, \end{cases} \tag{11}$$

and reduced to (10) otherwise.

The quantities we are looking for are, with the proper normalization

$$\psi(x,t) = \sum_{n=1}^{N} \sqrt{\omega_i^s} c_n(t) \Pi_n(x), \qquad \text{and} \qquad \frac{d}{dx}\psi(x,t) = \sum_{n=1}^{N} \sqrt{\omega_i^s} c_n(t) \frac{d}{dx}\Pi_n(x). \tag{12}$$

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The FEDVR basis - The nodes

$$N_s = 4,\ N_p = 5, \qquad N = 15,\ \ N' = 17.$$

$$n' = (s-1)(N_p - 1) + i$$

$$N = N_s(N_p - 1) - 1, \qquad N = 15$$

$$N' = N_s(N_p - 1) + 1 \qquad N' = 17.$$

- $\mathcal{B}'$ provides mathematical support
- $\mathcal{B}$ is where we have the expression of the wavefunction
- We note that $n' = n + 1$ when defined

		$\mathcal{B}'$				$\mathcal{B}$	
n'	s'	i'	$x_{n'}$	n	s	i	x_n
1	1	1	-5.0000000000000000				
2	1	2	-4.5683170883849709	1	1	1	-4.5683170883849709
3	1	3	-3.7500000000000000	2	1	2	-3.7500000000000000
4	1	4	-2.9316829116150291	3	1	3	-2.9316829116150291
5	1	5	-2.5000000000000000	4	1	4	-2.5000000000000000
5	2	1	-2.5000000000000000				
6	2	2	-2.0683170883849713	5	2	1	-2.0683170883849713
7	2	3	-1.2500000000000000	6	2	2	-1.2500000000000000
8	2	4	-0.43168291161502864	7	2	3	-0.43168291161502864
9	2	5	0.0000000000000000	8	2	4	0.0000000000000000
9	3	1	0.0000000000000000				
10	3	2	0.43168291161502864	9	3	1	0.43168291161502864
11	3	3	1.2500000000000000	10	3	2	1.2500000000000000
12	3	4	2.0683170883849713	11	3	3	2.0683170883849713
13	3	5	2.5000000000000000	12	3	4	2.5000000000000000
13	4	1	2.5000000000000000				
14	4	2	2.9316829116150291	13	4	1	2.9316829116150291
15	4	3	3.7500000000000000	14	4	2	3.7500000000000000
16	4	4	4.5683170883849709	15	4	3	4.5683170883849709
17	4	5	5.0000000000000000				

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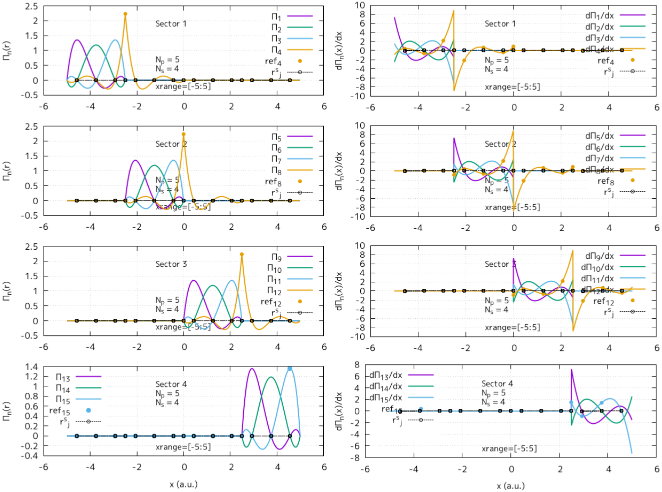
The FEDVR basis - Π_n and its derivative



$$\Pi_n(x) = \sum_{m=1}^{N} \Pi_n(x_m)\hat{\boldsymbol{e}}_m \qquad (13)$$

- The continuous line represent Eq. (5) for an arbitrary x in the sector's range
- The colored dot represents the value for the different nodes x_j, i.e. Eq. (13), with $\hat{\boldsymbol{e}}_m$ the unitary vectors of the cartesian basis

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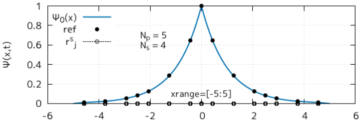
The FEDVR basis - The analytical wavefunction



$$\pi_i^s(r) = \frac{1}{\sqrt{\omega_i}} \prod_{\substack{k=1 \\ k\neq i}}^{N_p} \frac{r - r_k^s}{r_i^s - r_k^s}, \qquad \pi_i^s(r_k^s) = \delta_{ik}/\sqrt{\omega_i^s},$$

$$\Pi_n(r) = \frac{1}{\sqrt{\Omega_n}} \begin{cases} \sqrt{\omega_{N_p}^s}\,\pi_{N_p}^s(r), & r \in [r_{s-1}, r_s]\,, \\ \sqrt{\omega_1^{s+1}}\,\pi_1^{s+1}(r), & r \in [r_s, r_{s+1}]\,, \end{cases} \qquad \Pi_n(x_m) = \delta_{nm}/\sqrt{\Omega_n},$$

for $n = (s-1)(N_p-1) + N_p$ and reduces to $\pi_i^s(r)$ otherwise

- The black dots represent (14) for $m = 1, \ldots, N$
- The blue line represent (15) in the whole range

$$\psi(x_m, t) = \sum_{n=1}^{N} \sqrt{\omega_i^s}\, c_n(t) \Pi_n(x_m) = \sum_{n=1}^{N} \sqrt{\omega_i^s}\, c_n(t) \delta_{nm}/\sqrt{\Omega_n} \tag{14}$$

$$\psi(x, t) = \sum_{n=1}^{N} \sqrt{\omega_i^s}\, c_n(t) \Pi_n(x) \tag{15}$$

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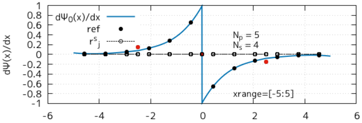
The FEDVR basis - The analytical derivative of the wavefunction

$$\left.\frac{d\pi_i^s(r)}{dr}\right|_{r=r_j^s} = \frac{1}{\sqrt{\omega_i}} \begin{cases} \frac{1}{r_i^s - r_j^s} \prod_{\substack{k=1 \\ k\neq i,j}}^{N_p} \frac{r_j^s - r_k^s}{r_i^s - r_k^s}, & i \neq j, \\ \frac{1}{2}\left(\left[\pi_i^s(r_{N_p}^s)\right]^2 - \left[\pi_i^s(r_1^s)\right]^2\right), & i = j, \end{cases}$$

$$\frac{d\Pi_n(r)}{dr} = \frac{1}{\sqrt{\Omega_n}} \begin{cases} \sqrt{\omega_{N_p}^s}\, d\pi_{N_p}^s(r)/dr, & r \leqslant r_s, \\ \sqrt{\omega_1^{s+1}}\, d\pi_1^{s+1}(r)/dr, & r \geqslant r_s, \end{cases} \quad \text{with} \quad \frac{d\pi_i^s(r)}{dr} = \frac{1}{\sqrt{\omega_i^s}} \sum_{\substack{j=1 \\ j\neq i}}^{N_p} \frac{1}{r_i^s - r_j^s} \prod_{\substack{k=1 \\ k\neq j,i}}^{N_p} \frac{r - r_k^s}{r_i^s - r_k^s},$$



for $n = (s-1)(N_p-1) + N_p$ and reduces to $d\pi_i^s(r)/dr$ otherwise

- The black dots represent (16) for $m = 1, \ldots, N$ (ref)
- The blue line represents (17) in the whole range

The values for $n' = (s-1)(N_p-1) + N_p$ are off

$$\frac{d\psi(x_m,t)}{dx} = \sum_{n=1}^{N} \sqrt{\omega_i^s} c_n(t) \frac{d\Pi_n(x_m)}{dx} \quad (16)$$

$$\frac{d\psi(x,t)}{dx} = \sum_{n=1}^{N} \sqrt{\omega_i^s} c_n(t) \frac{d\Pi_n(x)}{dx} \quad (17)$$

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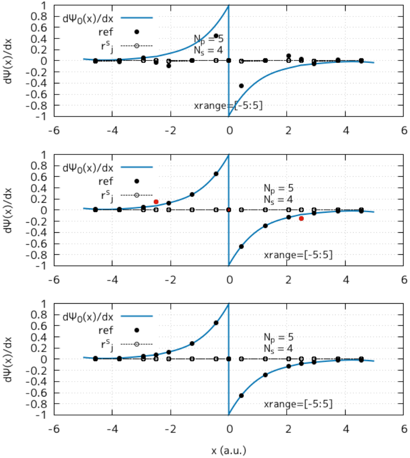
The FEDVR basis - Extension by continuity



- (Up) What was shown last week
- (Middle) Improvement of the discrete formula
- (Down) Fixing the values at the boundaries

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Propagators

- We are now interested in solving the following 1D problem with a ZRP

$$\begin{cases} i\frac{\partial}{\partial t}\psi(x,t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\psi(x,t), \\ i\frac{\partial}{\partial t}\phi_\omega(x,t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\phi_\omega(x,t) + e^{i\omega t}\hat{p}\,\psi(x,t), \end{cases} \tag{18}$$

- So we want our drivers to solve the following general case as efficiently as possible

$$i\partial_t\phi_\omega(x,t) = H(t)\phi_\omega(x,t) + s(t), \qquad H(t) = H_0 + F(t)x, \tag{19}$$

- Considering, the solution to the homogeneous problem, without the source

$$i\partial_t U(t,t_0) = (H_0 + F(t)x)\,U(t,t_0), \qquad U(t,t_0) = \mathbb{1}, \tag{20}$$

we can them, from Duhamel principle write

$$\partial_t\phi_\omega(x,t) = U(t,t_0)\phi_\omega(x,t) - i\int_{t_0}^{t_1} U(t-\tau)s(\tau)d\tau. \tag{21}$$

The above is the exact formulation of our problem.

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Propagators

- We are now interested in solving the following 1D problem with a ZRP

$$\begin{cases} i\frac{\partial}{\partial t}\psi(x,t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\psi(x,t), \\ i\frac{\partial}{\partial t}\phi_\omega(x,t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\phi_\omega(x,t) + e^{i\omega t}\hat{p}\,\psi(x,t), \end{cases} \tag{22}$$

- So we want our drivers to solve the following general case as efficiently as possible

$$i\partial_t\phi_\omega(x,t) = H(t)\phi_\omega(x,t) + s(t), \qquad H(t) = H_0 + F(t)x, \tag{23}$$

- Considering, the solution to the homogeneous problem, without the source

$$i\partial_t U(t,t_0) = (H_0 + F(t)x)\,U(t,t_0), \qquad U(t,t_0) = \mathbb{1}, \tag{24}$$

  we can them, from Duhamel principle write

$$\partial_t\phi_\omega(x,t) = U(t,t_0)\phi_\omega(x,t) - i\int_{t_0}^{t_1} U(t-\tau)s(\tau)d\tau. \tag{25}$$

  The above is the exact formulation of our problem.

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Time evolution

$$\partial_t \phi_\omega(x,t) = U(t,t_0)\phi_\omega(x,t) - i\int_{t_0}^{t_1} U(t-\tau)s(\tau)d\tau. \tag{25}$$

- (25), or (24) can be sold by any known solvers
- We will try, mainly
  1. Split-operator (Midpoint Duhamel)
  2. Adams methods
  3. Runge-Kutta 4 (IFRK4)
  4. etc...

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Propagators

$$\partial_t \phi_\omega(x,t) = U(t,t_0)\phi_\omega(x,t) - i\int_{t_0}^{t_1} U(t-\tau)s(\tau)d\tau. \tag{25}$$

- Our test system will be a forced QHO, with source

$$\begin{cases} i\partial_t \psi(x,t) & = \left(-(1/2)\partial_x^2 + (1/2)(x)^2 + F(t)x\right)\psi(x,t), \qquad (\hbar = \omega = m = 1) \\ i\partial_t \phi_\omega(x,t) & = \left(-(1/2)\partial_x^2 + (1/2)(x)^2 + F(t)x\right)\phi_\omega(x,t) + e^{i\omega t}\hat{p}\,\psi(x,t), \end{cases} \tag{26}$$

  1. We know how to evaluate the 1st term, order considerations set aside
  2. We will formally evaluate the 2nd term, firstly by
     - Split-operator (Midpoint Duhamel)
     - Adams methods
     - Runge-Kutta 4 (IFRK4)
     - etc...
- The 1st and 2nd term evaluations shall be done with matching order methods
- Because the test needs to conform the form of (Eq.(22)), it is extremely challenging to carry the absolute convergence test reliably (see the following slides)

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Convergence criteria

$$\partial_t y = Ay + S(x,t). \tag{27}$$

- We assess our propagation drivers based on two different tests
  - Absolute convergence

$$E_\ell^{\text{abs}} = \left| y_\ell(T) - y_{\text{exact}}(T) \right|, \tag{28}$$

$$p_\ell^{\text{abs}} = \frac{\log E_{\ell-1}^{\text{abs}} / E_\ell^{\text{abs}}}{\log(2)} \tag{29}$$

  - Self-convergence

$$D_\ell = \left| y_\ell(T) - y_{\ell-1}(T) \right|, \qquad D_\ell \sim C h_\ell^p \tag{30}$$

$$p_\ell^{\text{self}} = \frac{\log\left(D_{\ell-1}/D_\ell\right)}{\log(2)} \tag{31}$$

where the number of steps N, and the time step h are such as

$$N_\ell = N_0 2^\ell, \qquad h_\ell = \frac{T}{N_0 2^\ell}. \tag{32}$$

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Midpoint Duhamel

$$\partial_t \phi_\omega(x,t) = U(t,t_0)\phi_\omega(x,t) - i\int_{t_0}^{t_1} U(t-\tau)s(\tau)d\tau. \tag{25}$$

- We can show that the above can be expressed as

$$\phi_{n+1}^\omega = U(\Delta t)\phi_n^\omega - i\Delta t\, U\left(\Delta t/2\right) S_{n+1/2}. \tag{33}$$

where the source is evaluated at $t+\Delta t/2$, i.e

$$S_{n+1/2} = e^{i\omega(t+\Delta t/2)}\partial_x \psi_{n+1/2}, \qquad \text{with} \qquad \psi_{n+1/2} = U(\Delta t/2)\psi_n. \tag{34}$$

This is the midpoint Duhamel formula, of order 2.

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Midpoint Duhamel - Results

$$\phi_{n+1}^{\omega} = U(\Delta t)\phi_n^{\omega} - i\Delta t\, U\left(\Delta t/2\right) S_{n+1/2}. \tag{33}$$

$$S_{n+1/2} = e^{i\omega(t+\Delta t/2)}\partial_x \psi_{n+1/2}, \qquad \text{with} \qquad \psi_{n+1/2} = U(\Delta t/2)\psi_n. \tag{34}$$

t	$\Re\,\psi(1)$	$\Im\,\psi(1)$	$\Vert\psi\Vert$	$\Re\,\phi(1)$	$\Im\,\phi(1)$	$\Vert\phi\Vert$
−1.0	1.0000	−0.0000	1.0000	0.0574	−0.0000	0.6248
−0.9	0.9979	−0.0482	1.0000	0.0640	−0.0018	0.6432
−0.5	0.9377	−0.2263	1.0000	0.1193	−0.0066	0.7814
−0.1	0.7927	−0.3337	1.0000	0.1885	−0.0195	0.8795
0.0	0.7484	−0.3437	1.0000	0.2036	−0.0192	0.8823
0.5	0.5428	−0.2995	1.0000	0.2925	0.0198	0.8257
0.9	0.4566	−0.2195	1.0000	0.3965	0.0396	0.8201
1.0	0.4474	−0.2054	1.0000	0.4222	0.0347	0.8347

Refinement level k	Time step Δt	$\vert\psi^{(k)} - \psi^{(k+1)}\vert$	$\vert\phi^{(k)} - \phi^{(k+1)}\vert$	Observed order
1	.100	3.6596×10^{-6}	2.7621×10^{-5}	—
2	.050	9.1488×10^{-7}	4.9953×10^{-6}	—
3	.025	2.2872×10^{-7}	1.2522×10^{-6}	$p_\psi = 2.0000$ $p_\phi = 1.9961$

- Results of the time propagation for the coupled forced QHO with and without source (Eq.26)

$$\mathrm{Ne} = 2$$

$$\mathrm{Np} = 6$$

$$-\mathrm{xmin} = \mathrm{xrange}/2 = 7$$

$$-t_0 = \mathrm{duration}/2 = 1$$

Initial conditions

$$\psi_0(x) = \exp(-(1/2)(x - x_0)^2)$$

$$\phi_0^{\omega}(x) = \partial_x \psi_0(x)$$

$$x_c = 0$$

$$p_k^{\mathrm{self}} = \left(\mathrm{err}_{k-1}/\mathrm{err}_k\right)/\log(2)$$